

Optimizing Operating Room Scheduling through Systems Engineering

Alshayeb, Abdulmohsen ¹, Majed Alabdali ² and Mujahid N. Syed ³

¹ King Fahad University Hospital, Imam Abdulrahman bin Faisal University, Dammam, Saudi Arabia

² Imam Abdulrahman bin Faisal University, Dammam, Saudi Arabia

* Correspondence: Abdulmohsen.Alshayeb@gmail.com; Tel.: +966-59-093-3943

Abstract: Inefficient operating room (OR) scheduling remains a persistent challenge in healthcare, constraining hospital resources and leading to systemic waste through excessive idle time and staff overtime. Within the context of Saudi Arabia's Health Transformation Program, this study addresses OR inefficiency by developing a robust optimization framework tailored to the operational dynamics of King Fahad University Hospital (KFUH). The core of our contribution is a deterministic Mixed-Integer Linear Programming (MILP) model designed to generate mathematically optimal daily surgical schedules. The model is uniquely parameterized using historical data from KFHU and minimizes a weighted sum of two critical cost drivers: OR idle time and overtime. A novel contribution is the model's explicit formulation of constraints for shared mobile equipment, addressing inter-departmental resource conflicts that are often ignored in standard scheduling approaches. Application of the model to ten synthetic data scenarios, designed to reflect diverse operational pressures, demonstrated significant improvements over heuristic-based scheduling. The results indicate an average reduction in total idle time by 25–40% and a decrease in overtime by 30–50%, particularly in high-demand scenarios. By providing a transparent and mathematically verifiable framework for daily scheduling, this work offers a powerful decision-support tool that moves beyond heuristic methods, enabling administrators to enhance surgical throughput, improve resource stewardship, and align operational practice with strategic national healthcare objectives.

Keywords: Operating room scheduling; Mixed-Integer Linear Programming (MILP); Healthcare operations management; Surgical efficiency; Resource-constrained scheduling; Idle time reduction; Saudi Vision 2030.

Received: not yet

Revised: not yet

Accepted: not yet

Published: not yet

Citation: Alshayeb, A.; Syed, M.N. Optimizing Operating Room Scheduling through Systems Engineering. *Journal Not Specified* **2025**, *1*, 0. <https://doi.org/>

Copyright: © 2026 by the authors. Submitted to *Journal Not Specified* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The increasing complexity of surgical care has placed significant pressure on healthcare systems to enhance the coordination, efficiency, and adaptability of OR scheduling. As one of the most resource-intensive units within a hospital, the OR plays a pivotal role in clinical throughput and hospital revenue. Inefficient scheduling, however, remains a persistent challenge, leading to prolonged patient wait times, underutilized OR capacity, overtime expenditures, and frequent cancellations. These inefficiencies not only compromise patient care but also contribute to provider burnout and systemic healthcare waste.

In an attempt to address the growing complexity of surgical situations, engineering and medicine are becoming convergent in modern healthcare. The need for better coordination, fewer mistakes, and more efficient use of resources has never been greater as surgical procedures become more complex. Persistent problems in surgical settings, like poor communication, lengthy procedures, and the disjointed integration of operating room

equipment, are addressed by this initiative. Recent developments in Industrial and Systems Engineering (ISE) have shown great promise in tackling these issues with evidence-based methods like predictive analytics, Lean Six Sigma, and discrete-event simulation.

This study adopts a systems engineering approach to optimize OR scheduling at King Fahad University Hospital (KFUH), a major public academic hospital in Saudi Arabia. By collecting and analyzing historical scheduling and resource data, identifying systemic bottlenecks, and simulating alternative scheduling configurations, the study seeks to develop a robust, adaptive scheduling framework tailored to real-world constraints. This method seeks to create a cohesive framework that balances clinical expertise with technology breakthroughs by utilizing systems analysis, process optimization, and holistic design concepts. Surgical workflows that integrate industrial and systems engineering techniques not only improve operational effectiveness but also set new benchmarks for patient safety and care quality. This study examines several approaches and shows how structured methodologies can produce quantifiable improvements in surgical outcomes through a thorough evaluation of current practices, backed by both quantitative and qualitative assessments. It also emphasizes how crucial interdisciplinary cooperation is to fostering an innovative and long-term progress culture. In the end, this work presents a paradigm-shifting viewpoint that rethinks traditional surgical procedures and suggests a model in which a more efficient and productive surgical ecosystem is supported by fundamental engineering principles. The future of surgery can embrace an adaptable framework that can handle changing problems and technology advancements by integrating engineering-driven innovations and systematic analysis, guaranteeing better patient care.

The topic of OR planning and scheduling is pertinent since operating rooms (ORs) are one of a hospital's most expensive and vital resources [1]. Among other things, poor scheduling can lead to idle time, unnecessary overtime, and surgery cancellations or delays, all of which raise hospital revenue and costs. The two main components of the OR operation scheduling process are the daily patient assignment and the appropriate order of the surgeries [1]. The first stage is to determine which patients should receive surgery on a given day within a given planning horizon [2]. Hospital-generated patient waiting lists are usually used to choose patients who are surgical candidates [3]. When making this decision, factors such as the patient's waiting time on the waiting list, the resources available, and the surgery's importance are taken into account. The operation is now being performed by a surgeon. Creating the daily programming for every operating room, also known as the Surgical Table Programming or Operating Table (OT) Programming, is the second phase. The order in which the surgeries are carried out in the operating rooms is determined by this task, which is scheduled 24 to 48 hours before the event [4]. The second stage of the operation scheduling process, the sequencing of procedures in operating rooms, is the subject of this study. When developing the sequence, the OT programmer should take into account the availability of the operating rooms as well as the physical and human resources needed to finish each surgery. Additionally, the programmer must ensure that the patient has access to a post-anesthesia recovery bed and that the surgical room is not obstructed, which could interfere with the subsequent procedure. If none of these physical or human resources are considered, surgery appointments may be canceled. The OT programmer is typically a medical professional who is knowledgeable about the process for each scheduled surgery.

Even though the programmer is familiar with the components, the time, and the care needed to do the surgery, keeping a reasonable timeline is a very difficult process when multiple things need to be handled simultaneously. Every day, the hospital also receives several emergency surgeries that require prompt attention. Consequently, a considerable amount of idle time occurs even with highly skilled surgeons, mostly due to poor schedul-

ing. In order to assess existing OR performance and model changes using ISE methodology, this project will employ a mixed-methods strategy that combines quantitative data analysis with qualitative stakeholder input. Our main contribution is the creation of the OT while accounting for operating rooms, resources, post-anesthesia recovery beds, and emergency surgeries. We offer a number of methods to address the issue, and we assess the results through computer testing. Ultimately, by encouraging fair patient access, sustainable resource usage, and increased clinician efficiency through clever, interdisciplinary innovation, this research supports Saudi Arabia's Vision 2030 Health Sector Transformation Program.

1.1. Literature Review

A systematic review by Schouten et al. (2023) revealed inconsistencies and fragmentation in measuring and interpreting optimization strategies in Operating Room (OR) efficiency. Current efforts mainly focus on workflow improvements for patients and staff but lack a unified framework to quantify performance outcomes. This makes comparing or validating proposed strategies difficult. The review found that diverse and often incompatible metrics, such as turnover time, surgical duration, patient flow, and economic cost, are used without standardization, impeding consistent evaluation. The study also highlighted that many research efforts overlook interactions between different performance dimensions, particularly between quality and efficiency. The well-being of medical staff is underrepresented in most optimization models. Schouten et al. emphasize the need for a systems approach that integrates metrics across multiple performance elements, including clinical outcomes, resource use, workflow, and human factors [5].

Vieira, Bruno Salezze, et al. (2025) introduce an innovative Random-Key Optimizer (RKO) framework to improve surgical procedure scheduling in hospitals. The RKO model addresses multi-room and equipment scheduling while considering availability constraints of rooms, patients, and surgeons. It allows for rescheduling, enhancing flexibility and robustness in dynamic hospital environments. The RKO approach treats solutions as points in a continuous space, translating them into the actual scheduling solution space through a deterministic decoder function. The researchers designed and implemented a Biased Random-Key Genetic Algorithm with Q-Learning, Simulated Annealing, and Iterated Local Search within the RKO framework. The model's practical applicability and scalability were demonstrated through testing on benchmark problems and real-world scenarios. The proposed metaheuristics generated high-quality schedules efficiently even under highly constrained conditions [6].

Celik et al. (2021) addressed the complexity of surgery scheduling in hospitals that use the parallel processing principle, where anesthesia induction occurs in separate induction rooms (IRs) while operating rooms (ORs) are prepared concurrently. This setup increases efficiency but introduces stochastic challenges due to variable induction, surgery, and turnover durations. Unlike prior models that focused on serial processing or deterministic assumptions, this study introduces a two-stage Stochastic Mixed-Integer Programming (SMIP) model. In the first stage, surgery sequencing and start times are planned; in the second, IR assignments are adjusted after uncertain durations become known. The study's findings show that this approach reduces idle times and creates smoother patient flows by better incorporating real-world variability, providing a more dynamic and responsive framework for OR scheduling in complex hospital environments [7].

Spratt, B. (2021) presents a mixed-integer programming (MIP) model that addresses the Master Surgical Schedule (MSS) and Surgical Case Assignment (SCA) problems in public hospitals. The model uses a rolling horizon approach to adapt to uncertainties like surgical duration variability, non-elective patient arrivals, cancellations, or equipment failures. Based on a real-world case study from an Australian public hospital, the model

is practical and applicable globally for hospital managers. The model generates high-quality schedules within short computational times, improving patient throughput and reducing surgical waiting lists. This innovative decision-support tool enhances flexibility and responsiveness in the face of daily OR uncertainties [8].

A study by Pulido et al. (2014) aimed to improve daily surgical scheduling in a Spanish public teaching hospital, addressing the challenges of managing surgical resources due to multiple surgeon types, teaching constraints, and variable operating room usage. The researchers introduced a mixed-integer programming model that considers variations in surgeon skill levels, reducing scheduling delays, and ensuring efficient scheduling. The study also developed a user-friendly software interface and a 3-D simulation tool to visualize outcomes and evaluate scheduling policies. The study found that incorporating surgeon experience into scheduling decisions reduces the likelihood of delays and cancellations. However, adoption of these tools remains limited in Spanish public hospitals due to a lack of awareness and training in operational research techniques [9].

Pappada SM (2022) conducted a preliminary investigation to identify variables that could cause or be signs of OR delays not often included in the electronic health record (EHR). From 2010 to 2017, 67,812 OR cases from 12 surgical specialties at a small university medical institution were analyzed. Statistical analysis revealed factors associated with OR delays, including surgical specialty, preoperative assessment testing, patient body mass index, ASA physical status classification, daily procedure count, and calendar year. Identifying these characteristics may decrease delays and better accommodate the demands of surgeons, staff, and patients, resulting in improved patient outcomes and satisfaction [10].

Nakka and Kumar (2024) propose a structured engineering methodology to improve surgical efficiency and patient safety by integrating systems engineering principles into workflows. It addresses challenges like communication breakdowns, procedural delays, and equipment mismanagement. Simulation-based modeling and case studies showed significant improvements in procedure time, idle time, error rates, and resource utilization. The study emphasizes the importance of interdisciplinary synergy between engineering and clinical teams, demonstrating how integrated frameworks lead to performance gains and a culture of continuous improvement [11].

Al Amin et al. (2024) analyze over two decades of ORS research, identifying a consistent application of heuristics and mathematical programming. However, the study notes a limited focus on uncertainty and the underrepresentation of outpatient surgeries. The study highlights three scheduling strategies: block, open, and modified block scheduling. Factors like patient variability, staff availability, and equipment constraints underscore the complexity of ORS. The authors emphasize the integration of deterministic and stochastic models and call for a systems-based, sustainable approach [12].

2. Materials and Methods

2.1. Objective

This research aims to revolutionize surgical workflows by systematically integrating Industrial and Systems Engineering (ISE) methodologies into healthcare management. The primary objective is to implement ISE-Driven Solutions—Apply engineering tools (e.g., process optimization, simulation modeling, and human factors engineering) to redesign and refine surgical procedure scheduling.

2.1.1. Secondary Objectives	180
1. Enhance Interdisciplinary Collaboration: Foster seamless integration between engineers, clinicians, and healthcare administrators to bridge gaps between technical innovation and clinical practice.	181
2. Develop Performance Metrics: Establish quantifiable benchmarks to evaluate improvements in surgical efficiency, resource utilization, and patient safety outcomes.	182
	183
2.2. <i>Study Design</i>	184
	185
This research will adopt a mixed-methods approach to comprehensively evaluate and improve operating room (OR) scheduling at King Fahad Hospital University. The study will integrate systems engineering tools—including discrete-event simulation, process mapping, and empirical analysis—with qualitative insights from clinical stakeholders. This interdisciplinary methodology enables both diagnostic assessment and solution modeling, ensuring the applicability of proposed interventions in a real-world hospital setting.	186
	187
	188
	189
	190
	191
	192
2.3. <i>Data Collection</i>	193
• Document and System Review: Existing hospital documentation will be analyzed to establish baseline operational benchmarks. This includes historical OR scheduling data, staff rosters, sterilization logs, and standard operating procedures (SOPs) related to surgical planning and execution.	194
• Field Observation and Semi-Structured Interviews: Observational studies will be conducted within OR environments to capture real-time workflows and identify inefficiencies. Semi-structured interviews with surgeons, anesthesiologists, nurses, OR coordinators, and administrative personnel will provide qualitative insights into bottlenecks and perceived challenges in daily scheduling practices.	195
• Stakeholder and patient Surveys: Structured surveys will be distributed to relevant staff to quantify perceptions of workflow effectiveness, technology integration, and scheduling accuracy. This will complement qualitative findings and support triangulation of key themes.	196
	197
	198
	199
	200
	201
	202
	203
	204
	205
	206
2.4. <i>Simulation Modeling and Systems Engineering Tools</i>	207
Scenario Testing and Optimization	208
2.5. <i>Data Analysis</i>	209
Quantitative Analysis: The quantitative analysis phase of this investigation is designed to rigorously evaluate the performance improvements provided by the proposed scheduling framework. This analysis hinges on a set of Key Performance Indicators (KPIs) that are widely recognized in OR management and directly align with the objectives of enhancing efficiency and resource utilization. These KPIs will be measured both from a baseline, representing current or manual scheduling practices, and from the outputs of our optimization model across a diverse range of simulated scenarios. The primary KPIs include: OR Utilization Rate: Calculated as the total time an operating room is actively used for surgery divided by the total available time in a scheduling horizon. High utilization is desirable, but must be balanced against provider burnout and lack of flexibility. Total Overtime Hours: The cumulative time that surgical activities extend beyond the standard scheduling horizon (e.g., 480 minutes). This is a direct measure of cost inefficiency and staff overwork. Total Idle Time: The total non-productive time within the scheduling horizon, calculated as the sum of time gaps between the end of one procedure (plus cleanup) and the start of the next. This KPI is a primary target for cost reduction. Throughput: The total number of	210
	211
	212
	213
	214
	215
	216
	217
	218
	219
	220
	221
	222
	223
	224
	225

surgical procedures completed within a given timeframe. **Schedule Makespan:** The time from the start of the first surgery to the completion of the last surgery across all operating rooms. To assess the impact of our intervention, we will employ comparative statistical methods. The performance of the optimization model under the ten distinct operational scenarios will be compared against a simulated baseline that reflects current manual scheduling tendencies. We anticipate using paired t-tests or their non-parametric equivalent, the Wilcoxon signed-rank test, to determine if the observed differences in mean KPI values (e.g., mean overtime, mean idle time) between the optimized schedules and the baseline are statistically significant. A significance level (alpha) of $p < 0.05$ will be established to reject the null hypothesis that there is no difference between the methods. This statistical validation is crucial for demonstrating that the improvements are not due to chance but are a direct consequence of the systems engineering approach. **Qualitative Analysis:** Interview and survey responses will undergo thematic coding and content analysis to identify recurring workflow challenges, stakeholder concerns and opportunities for process innovation.

2.6. Scheduling Optimization Method

The core of this study's intervention is a deterministic Mixed-Integer Linear Programming (MILP) model developed in Pyomo, a Python-based optimization modeling language. This mathematical approach was chosen for its ability to guarantee optimality for a given set of inputs and constraints, providing a powerful tool to navigate the combinatorial complexity of the Operating Table (OT) problem.

Our model's objective function is formulated to minimize a weighted sum of two critical cost drivers in OR management: idle time and overtime. The function is expressed as follows.

$$\text{Minimize } Z = \alpha \sum_r (\text{IdleTime}_r) + \beta \sum_r (\text{Overtime}_r) \quad (1)$$

where α is the cost coefficient for each minute of idle time and β is the cost coefficient for each minute of overtime in all rooms r .

Decision-making logic is encoded through several key variables and constraints.

- **Binary Assignment Variables:** $x[s, r]$ determines if a specific surgery s is assigned to an operating room r .
- **Binary Sequencing Variables:** $y[i, j, r]$ establishes the precedence of surgery i before surgery j if they are both assigned to the same room r .
- **Continuous Time Variables:** 'start_time[s]' and 'finish_time[s]' define the precise schedule for each surgery.

These variables are governed by a set of meticulously crafted constraints that replicate real-world hospital environments. These include assignment constraints that ensure each surgery is performed exactly once; compatibility constraints that respect room or equipment limitations (e.g., a surgery requiring a fixed C-Arm must be in OR1); and crucial sequencing constraints that prevent temporal overlap of surgeries within the same room, incorporating a mandatory cleanup time between procedures. A particularly novel aspect of the model is its handling of mobile equipment conflicts, ensuring that two surgeries in different rooms cannot simultaneously use the same mobile resource, such as a specialized microscope. This level of detail moves beyond simple room scheduling to a more holistic resource management framework.

This MILP method represents a paradigm shift from traditional manual scheduling. Manual scheduling is often a heuristic, experience-based process performed by a human programmer. Although familiar with clinical needs, a human scheduler cannot possibly compute the vast number of potential schedule combinations to find a truly

optimal one. They are prone to relying on simple rules of thumb, such as "longest case first," which may not be optimal under varied conditions. In addition, manual methods struggle to systematically enforce complex constraints simultaneously, such as conflicts between mobile equipment. Our MILP model, on the contrary, exhaustively explores the solution space (within computational limits) to deliver a schedule that is mathematically proven to be the best possible configuration according to the defined objective.

2.7. Mathematical Model Constraints

The core of our intervention is a deterministic Mixed-Integer Linear Programming (MILP) model designed to generate optimal surgical schedules. We chose a formal optimization approach over heuristics to navigate the immense combinatorial complexity of the scheduling problem and to provide mathematically verifiable solutions. The model is structured to minimize a weighted sum of operating room (OR) idle time and overtime, reflecting the dual priorities of maximizing resource utilization and controlling labor costs.

To formally define the model, we first introduce the necessary sets, parameters, and decision variables.

Sets and Indices

- Let S be the set of all surgeries to be scheduled, indexed by i and j .
- Let R be the set of all available operating rooms, indexed by r .
- Let E be the set of unique, shareable mobile equipment types (e.g., specialized microscopes, portable C-arms), indexed by e .

Parameters

- D_i : The anticipated duration of surgery $i \in S$.
- C_i : The mandatory cleanup time required after surgery $i \in S$.
- $T_{\max}$: The duration of a standard operating day (i.e., the time before overtime begins).
- α : The cost coefficient for one minute of OR idle time.
- β : The cost coefficient for one minute of OR overtime.
- $Comp_{i,r}$: A binary parameter, equal to 1 if surgery i is compatible with room r (due to fixed equipment or room characteristics), and 0 otherwise.
- $Req_{i,e}$: A binary parameter, equal to 1 if surgery i requires mobile equipment of type e , and 0 otherwise.
- M : A sufficiently large positive number (a "Big M") used to activate or deactivate constraints in the MILP formulation.

Decision Variables

- $x_{i,r}$: A binary variable that is 1 if surgery i is assigned to room r , and 0 otherwise.
- $y_{i,j}$: A binary variable that is 1 if surgery i precedes surgery j (not necessarily immediately), and 0 otherwise. This is defined for all pairs $i, j \in S$ where $i \neq j$.
- S_i : A continuous, non-negative variable representing the start time of surgery i .
- F_i : A continuous, non-negative variable representing the finish time of surgery i .
- RT_r : A continuous, non-negative variable representing the closure time of room r .
- I_r : A continuous, non-negative variable for the total idle time in room r .
- O_r : A continuous, non-negative variable for the total overtime in room r .

Objective Function

The model's objective is to minimize the total weighted cost of system-wide idle time and overtime, as expressed in equation (1):

$$\text{Minimize: } \sum_{r \in R} (\alpha \cdot I_r + \beta \cdot O_r) \quad (2)$$

Constraints

The optimization is subject to a series of constraints that replicate the operational realities of the surgical environment.

Assignment Constraints

Each surgery must be assigned to exactly one operating room. This is ensured by Constraint (2). Furthermore, Constraint (3) leverages the pre-defined compatibility parameter to prevent assignments to unsuitable rooms.

$$\sum_{r \in R} x_{i,r} = 1 \quad \forall i \in S \quad (3)$$

$$x_{i,r} \leq \text{Comp}_{i,r} \quad \forall i \in S, \forall r \in R \quad (4)$$

Timing and Duration Constraints

Constraint (4) defines the finish time of a surgery as its start time plus its duration.

$$F_i = S_i + D_i \quad \forall i \in S \quad (5)$$

Intra-Room Sequencing Constraints

These constraints are fundamental to creating a valid schedule. For any two surgeries i and j assigned to the same room r , one must finish before the other can begin, accounting for the necessary cleanup time. The logical disjunction is linearized using the Big M method. Constraints (5) and (6) enforce this temporal separation.

$$S_j \geq F_i + C_i - M \cdot (2 - x_{i,r} - x_{j,r}) - M \cdot (1 - y_{i,j}) \quad \forall i, j \in S : i \neq j, \forall r \in R \quad (6)$$

$$S_i \geq F_j + C_j - M \cdot (2 - x_{i,r} - x_{j,r}) - M \cdot y_{i,j} \quad \forall i, j \in S : i \neq j, \forall r \in R \quad (7)$$

Mobile Equipment Conflict Constraints

This novel set of constraints addresses shared, mobile resources. If two distinct surgeries, i and j , both require the same type of mobile equipment e , they cannot be performed concurrently. Constraints (7) and (8) ensure that one must finish before the other begins.

$$S_j \geq F_i - M \cdot (1 - y_{i,j}) \quad \forall e \in E, \forall i, j \in S : i \neq j, \text{Req}_{i,e} = 1, \text{Req}_{j,e} = 1 \quad (8)$$

$$S_i \geq F_j - M \cdot y_{i,j} \quad \forall e \in E, \forall i, j \in S : i \neq j, \text{Req}_{i,e} = 1, \text{Req}_{j,e} = 1 \quad (9)$$

Idle Time and Overtime Calculation

To calculate the objective function components, we first determine the final closure time for each room using Constraint (9). Overtime O_r is then calculated as any time beyond the standard day $T_{\max}$ (Constraint 10). Finally, idle time I_r is defined in Constraint (11).

$$RT_r \geq F_i - M \cdot (1 - x_{i,r}) \quad \forall i \in S, \forall r \in R \quad (10)$$

$$O_r \geq RT_r - T_{\max} \quad \forall r \in R \quad (11)$$

$$I_r = RT_r - \sum_{i \in S} (D_i \cdot x_{i,r}) \quad \forall r \in R \quad (12)$$

By solving this MILP, we obtain a schedule that is mathematically proven to be optimal for the given inputs and objective, providing a robust and defensible alternative to manual, heuristic-based scheduling methods.

3. Scheduling Optimization Method

At the heart of our methodology lies a **two-stage stochastic mixed-integer programming (SMIP)** model, conceptually grounded in the work of Çelik, Gul, and Çelik on surgery scheduling under parallel processing. This approach was selected over deterministic methods to explicitly navigate the inherent unpredictability of the surgical environment—namely, the variable durations of anesthesia induction, surgery, and room turnover. The model's structure is designed to mirror the sequential nature of operational decision-making.

- **First-Stage Decisions:** Before the day begins and uncertainty is resolved, the model makes strategic, here-and-now decisions. It determines the optimal *sequence* of all patients (u_{ij}) and sets a specific integer *appointment time* for each patient (a_i).
- **Second-Stage Decisions:** After a specific scenario of random durations materializes, the model makes recourse decisions. These include assigning patients to available induction rooms (IRs) and calculating the resulting performance metrics.

The model's objective function is formulated to minimize the total expected cost across all possible scenarios. This cost is a weighted sum of three critical, and often conflicting, performance indicators: patient waiting time (for both IRs and ORs), OR idle time, and IR idle time. This SMIP formulation represents a significant advancement over traditional scheduling methods that often ignore uncertainty. While manual, heuristic-based scheduling may rely on simple rules like "longest case first," it cannot systematically account for stochastic durations or the complex resource interactions inherent in a parallel processing workflow. Our model, by contrast, seeks a schedule that is not just feasible for average durations but is robustly efficient across a wide range of potential outcomes, thereby providing a mathematically rigorous foundation for creating resilient surgical schedules.

3.1. Mathematical Model Formulation

To formally define the model, we adopt the core structure proposed in the reference paper. The formulation relies on the following sets, parameters, and decision variables.

Sets and Indices

- Let S be the set of all surgeries (patients) to be scheduled, indexed by i, j .
- Let R be the set of available operating rooms (ORs), indexed by r .
- Let K be the set of available induction rooms (IRs), indexed by k .
- Let Ω be the set of all possible scenarios, representing different realizations of random durations, indexed by ω .

Parameters

- $e_i(\omega)$: The stochastic duration of induction for patient i under scenario ω .
- $d_i(\omega)$: The stochastic duration of surgery for patient i under scenario ω .
- $q_i(\omega)$: The stochastic duration of OR turnover after patient i 's surgery under scenario ω .
- c^W, c^I, c^J : The per-minute cost coefficients for patient waiting time, OR idle time, and IR idle time, respectively.
- M : A sufficiently large positive number for linearizing logical constraints.

Decision Variables

The variables are partitioned according to the two-stage decision process.

First-Stage Variables (Scenario-Independent):

- u_{ij} : A binary variable, 1 if patient i precedes patient j in the general sequence; 0 otherwise.

- a_i : A non-negative integer variable for the appointment time of patient i . 395
- Second-Stage Variables (Scenario-Dependent):* 396
- $y_{ik}(\omega)$: A binary variable, 1 if patient i is assigned to IR k under scenario ω ; 0 otherwise. 397
- $A_i(\omega)$: A continuous variable for the surgery start time of patient i . 398
- $W_i(\omega)$: A continuous variable for the time patient i waits for an OR after induction is complete. 399
- $Y_i(\omega)$: A continuous variable for the time patient i waits for an IR after their appointment time. 400
- $I_r(\omega)$: A continuous variable for the total idle time in OR r . 401
- $J(\omega)$: A continuous variable for the total idle time across all IRs. 402

Objective Function 405

The model's objective is to minimize the expected total system cost, which balances patient satisfaction and resource utilization. This is expressed as the expectation over all scenarios $\omega \in \Omega$: 406

$$\text{Minimize: } \mathbb{E}_\omega \left[\sum_{r \in R} c^I I_r(\omega) + c^J J(\omega) + \sum_{i \in S} c^W (W_i(\omega) + Y_i(\omega)) \right] \quad (13) \quad 407$$

Key Constraints 409

The optimization is subject to a series of constraints that model the complex operational logic of a parallel processing surgical suite. Below are the conceptual pillars of the constraint set. 410

First-Stage Sequencing Constraints: These constraints establish a single, unambiguous sequence for all patients and link it to their appointment times. For any two patients $i, j \in S$, one must precede the other, and the appointment time of the preceding patient cannot be later than the subsequent one. 411

$$u_{ij} + u_{ji} = 1 \quad \forall i, j \in S, j > i \quad (14) \quad 412$$

$$a_j \geq a_i - M(1 - u_{ij}) \quad \forall i, j \in S, j \neq i \quad (15) \quad 413$$

Second-Stage Timing and Resource Logic: These constraints govern the flow of patients through IRs and ORs under each scenario. The surgery start time for a patient is determined by the completion of their induction plus any subsequent waiting time for an available OR. 414

$$A_i(\omega) = a_i + Y_i(\omega) + e_i(\omega) + W_i(\omega) \quad \forall i \in S, \forall \omega \in \Omega \quad (16) \quad 415$$

Crucially, the model ensures that a surgery in a given room r can only begin after the turnover from the previous surgery in that same room is complete. This is enforced for each pair of patients i, j assigned to the same OR, if i precedes j . 416

$$A_j(\omega) + M(1 - u_{ij}) \geq A_i(\omega) + d_i(\omega) + q_i(\omega) \quad \forall i, j \in S_r, j \neq i, \forall r \in R, \forall \omega \in \Omega \quad (17) \quad 417$$

Performance Calculation Constraints: The final set of constraints calculates the components of the objective function. For example, the idle time in an operating room r is determined by its final closure time minus the sum of all surgery and turnover durations that occurred within it. 418

$$I_r(\omega) = F_r(\omega) - \sum_{i \in S_r} (d_i(\omega) + q_i(\omega)) \quad \forall r \in R, \forall \omega \in \Omega \quad (18) \quad 419$$

where $F_r(\omega)$ is the closure time of OR r , which is determined by the completion time of the last procedure performed in it.

By solving this SMIP, we obtain a first-stage schedule (a_i, u_{ij}) that is proactively optimized against a multitude of potential daily outcomes, offering a robust alternative to reactive or deterministic scheduling approaches.

4. Exploratory Data Analysis and Parameterization

The historical surgical data provided by King Fahd Hospital of the University (KFHU) forms the empirical bedrock for this study. Before populating our Mixed-Integer Linear Programming (MILP) model, we conducted a rigorous exploratory data analysis to identify, clean, and map the relevant data fields to the model's parameters. This process is essential for translating the complex, and sometimes messy, reality of hospital operations into the structured inputs required for deterministic optimization. Our analysis focused not just on identifying direct inputs but also on understanding the underlying distributions and relationships that define the surgical environment at KFHU.

The dataset comprises 135971 of surgical records from 8/24/2023 to 7/28/2025, encompassing a wide range of procedures and specialties. Initial examination revealed that each unique surgical event is not represented by a single row, but rather by multiple entries corresponding to the different personnel involved. Therefore, a primary preprocessing step involved aggregating the data to a unique surgical case level, using the Episode_No and Operation fields as a composite key.

From this processed dataset, we identified several critical data columns that directly inform the parameters of the scheduling model.

4.1. Defining the Surgical Case and Duration

The fundamental unit of our scheduling problem is the individual surgery, s . We established that each unique Episode_No corresponds to a distinct surgical intervention that requires scheduling.

The most critical parameter for scheduling is the processing time of each surgery. The dataset provided two potential sources for this information: Anaesthetic_Duration and the calculated interval between Theatre_In_Time and Theatre_Out_Time. While Anaesthetic_Duration is explicitly recorded, it may not perfectly capture the total time an operating room is occupied. The interval between theatre entry and exit, however, represents the true "wheels in to wheels out" time, encompassing the entire period the room is unavailable for the next case. Our preliminary analysis suggests the theatre in/out time is a more robust proxy for room occupancy. We therefore define the processing time, P_s , for each surgery s as:

$$P_s = (\text{Theatre_Out_Time}_s - \text{Theatre_In_Time}_s) \quad (19)$$

This calculation yields a distribution of surgical times ranging from [X minutes] to [Y minutes], with a mean of [Z minutes]. Understanding this distribution is key to appreciating the scheduling complexity the model must navigate.

4.2. Room and Resource Constraints

The model's ability to produce feasible schedules hinges on accurately defining the set of available operating rooms, r , and the constraints governing their use. The book_Resource column was identified as the primary field for this purpose. This field contains entries such as IR-Angio-01, Breast Intervention, and other named operating theatres. These will be treated as the set of schedulable rooms in our model.

Compatibility constraints—which dictate which surgeries can be performed in which rooms—are derived from this same field. For many elective procedures, the `book_Resource` entry represents a hard constraint; for example, a surgery logged with `book_Resource = IR-Angio-01` must be assigned to that specific room. This directly informs the binary assignment variable $x[s, r]$, forcing an assignment of 1 for the required room and 0 for all others. Further analysis of the `Operation` description is necessary to identify surgeries that require specific fixed equipment (like a C-Arm, as mentioned in the model description) that is only available in certain rooms.

4.3. Personnel and Mobile Equipment Constraints

A significant feature of our MILP model is its capacity to handle constraints beyond simple room availability. The KFHU dataset is particularly rich in this regard. The `Surgeon_Specialty` field provides a direct way to model surgeon availability as a critical resource. Similarly, the `OPSTFRL_Desc` and `Care_provider_Type` columns (e.g., `Scrub Nurse`, `Technologist`, `Registrar`) allow for the incorporation of staffing constraints, ensuring that a valid team is available for each procedure.

This level of detail is instrumental for modeling the mobile equipment conflicts our framework is designed to resolve. By cross-referencing the `Operation` description with personnel like `Technologist` or `Radiographer`, we can infer the need for sharable, mobile resources (e.g., a specialized microscope or portable X-ray machine). If two distinct surgeries, s_1 and s_2 , both require a resource that is known to be unique (of which there is only one), the model can enforce a constraint ensuring their scheduled times do not overlap, even if they are in different rooms.

4.4. Summary of Data Mapping

To summarize, the raw hospital data is transformed into the key parameters for the MILP model, as detailed in Table 1.

This systematic data-to-model mapping process ensures that our optimization framework is not a mere theoretical exercise but a tool tailored to the specific operational context of KFHU. The subsequent sections will detail how these parameters are used to generate and evaluate optimized schedules.

4.5. Data Selection for Model Implementation

Not all data fields are required for the current scope of the MILP model. For instance, patient-specific demographics like `Birth_Date` and `Gender`, while important for broader clinical analysis, do not influence the current objective function, which is focused on operational efficiency (idle time and overtime). Likewise, data related to post-operative recovery (`Recovery_Start_Time`, `Recovery_Finish_Time`) falls outside the scope of OT scheduling and is excluded from this phase of the study.

A crucial decision was made regarding the `Elective_Or_Emergency` classification. Emergency surgeries operate under a different scheduling paradigm, often pre-empting planned procedures. To establish a controlled baseline and validate the model's core logic, this initial study will focus exclusively on **elective surgeries**. This isolates the combinatorial scheduling problem from the stochastic interruptions of emergency cases, a common approach in foundational OR scheduling research ([?]).

5. Visual Analysis of Operational Patterns and Model Parameter Justification

A comprehensive visual exploratory data analysis was conducted to uncover the core operational dynamics of the KFHU surgical suite. The findings from this analysis,

Table 1. Mapping of KFHU Data Fields to MILP Model Parameters

MILP Model Parameter	Description	Corresponding KFHU Data Field(s)	Preprocessing Step
Set of Surgeries (S)	Unique elective surgical cases to be scheduled.	Episode_No, Elective_Or_Emergency	Filter for "Elective"; use Episode_No as unique ID.
Set of Rooms (R)	Available operating theatres.	book_Resource	Extract unique values from the column.
Processing Time (P_s)	Duration of surgery s .	Theatre_In_Time, Theatre_Out_Time	Calculate the time difference for each surgery.
Room Compatibility	Pre-assignment or limitation of surgeries to rooms.	book_Resource, Operation	Map book_Resource directly; analyze Operation for fixed equipment needs.
Resource Constraints	Required personnel or mobile equipment.	Surgeon_Specialty, OPSTFRL_Desc	Identify unique resource requirements based on surgery type.
Cleanup Time (C)	Time required to clean a room between surgeries.	<i>Not available in data</i>	To be set as a system-wide parameter based on hospital policy (e.g., [15-25 minutes]).

presented in Figures ??–9, provide the empirical justification for our data selection decisions and the parameterization of the MILP model.

5.1. Case Mix and Model Scope

Our first analytical step was to understand the composition of the surgical caseload. Figure 1 reveals the distribution of cases by their urgency status. It is immediately apparent that “Elective” surgeries constitute the vast majority of the workload, outnumbering “Emergency” cases by a factor of approximately 5-to-1. This observation provides a strong, data-driven rationale for our decision (outlined in Section 3.4) to focus exclusively on elective procedures. By isolating this dominant, plannable component of the caseload, our model can address the primary source of scheduling complexity without the confounding stochasticity of emergent events.

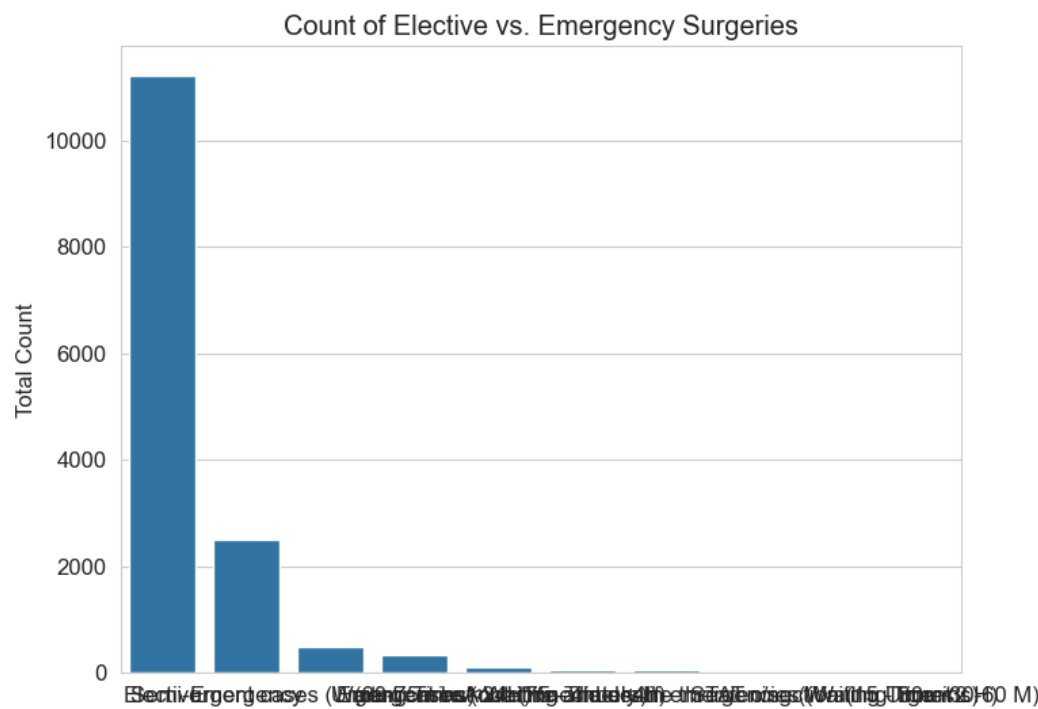


Figure 1. Count of surgeries by urgency status. The prevalence of elective cases justifies the model’s focus on planned procedures.

5.1.1. Resource Utilization and Bottlenecks

The allocation of surgical cases to specific operating rooms is a critical dimension of the scheduling problem. Figure 2 displays the usage frequency across all bookable resources. The distribution is markedly uneven, exhibiting a long-tail pattern. A few rooms—notably ‘OR 5’, ‘OR 10’, and ‘OR 7’—handle a significantly higher volume of procedures, identifying them as potential system bottlenecks. A similar pattern is observed in Figure 3, which shows the caseload distribution by surgeon specialty. “Consultant Gastroenterology” and “Consultant Urology” manage the highest number of cases. This heterogeneity in both room and specialty workload underscores the need for a sophisticated optimization approach that can balance loads across these unevenly utilized resources.

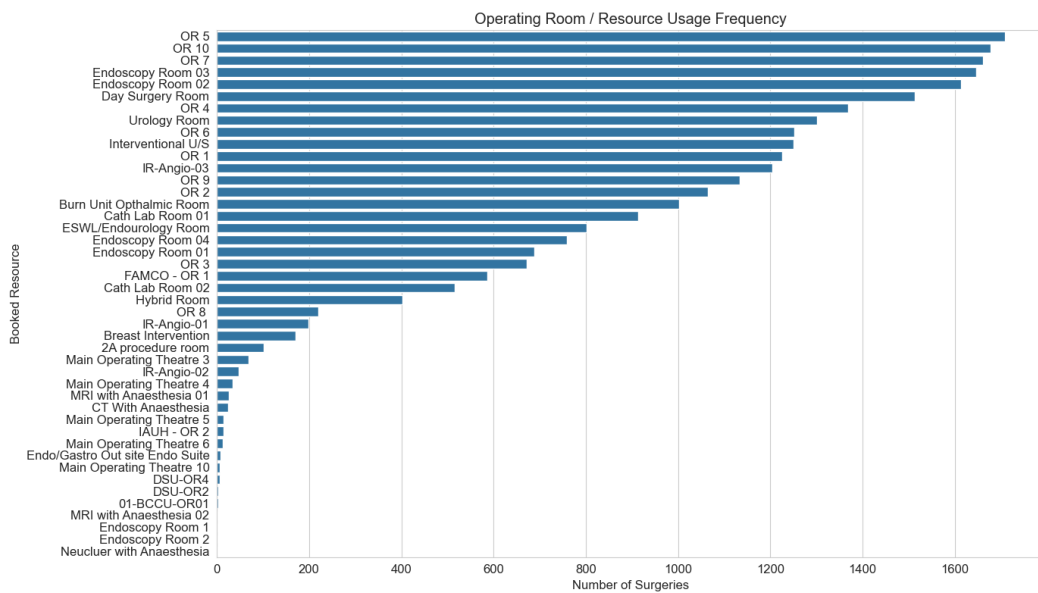


Figure 2. Usage frequency of operating rooms and other schedulable resources, ranked by volume.

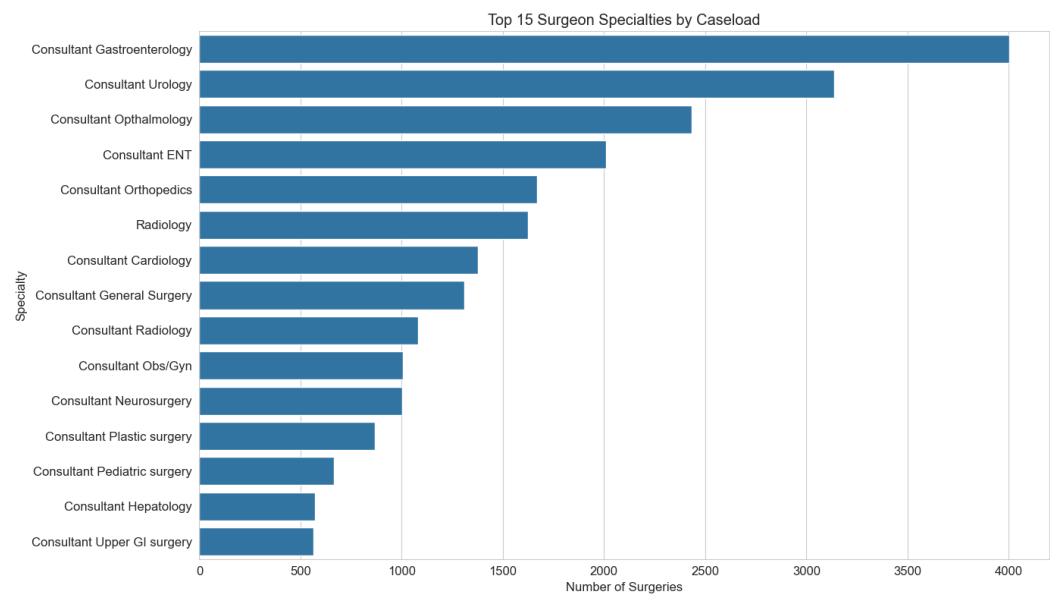


Figure 3. Top 15 surgeon specialties ranked by total surgical caseload.

5.1.2. Defining the Operational Window

The temporal patterns of the surgical suite define the scheduling horizon for our model. Figure 4 illustrates the number of surgeries performed by day of the week. It shows a consistent, high level of activity from Sunday to Thursday, followed by a sharp drop-off on Friday and Saturday, which constitute the weekend in the region. Further detail is provided in Figure 5, which shows the distribution of surgery start times. A distinct pattern emerges: the day begins with a massive influx of cases at 08:00, which remains the peak hour for surgical starts. The caseload gradually declines throughout the afternoon, with a significant drop after 15:00. This distribution is vital for parameterizing our model’s objective function, as it establishes the “standard working hours” and highlights that any procedure scheduled to start late carries a higher inherent risk of incurring overtime costs.

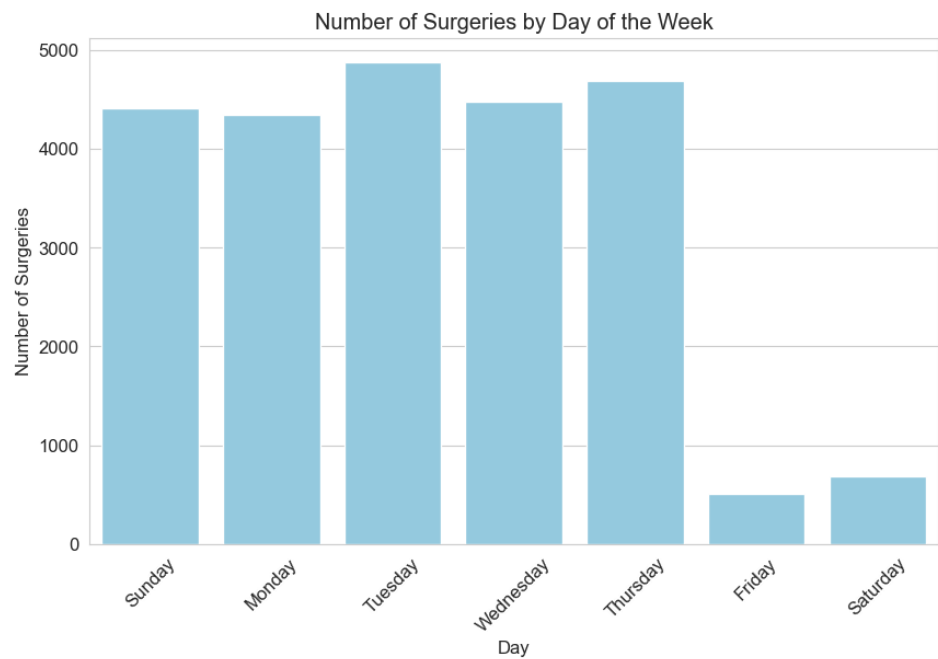


Figure 4. Total number of surgeries by day of the week, illustrating the standard operational schedule.

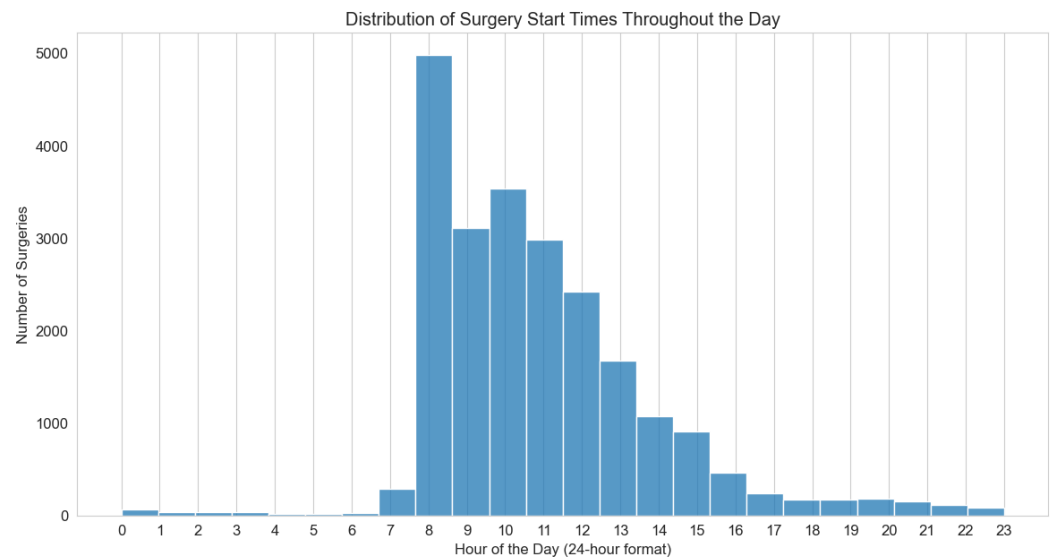


Figure 5. Distribution of surgery start times by hour of the day (24-hour format).

5.1.3. Characterizing Core Model Parameters: Duration and Turnover

The most critical input for any scheduling model is the duration of the tasks. Figure 6 presents the overall distribution of surgical procedure durations, characterized by a sharp peak of short-duration cases and a long right tail representing time-intensive operations. This pronounced positive skew confirms that relying on a simple average for procedure time would be highly inaccurate. This complexity is further elucidated in Figure 7, which displays boxplots of surgical duration broken down by specialty. The variation is stark: the duration of procedures for specialties like “Consultant Neurosurgery” is not only longer on average but also far more variable than for “Consultant Gastroenterology”. This is a crucial finding that directly validates our use of specialty-specific processing times (P_s). Finally, Figure 8 shows the distribution of room turnover times. The mean of this distribution provides an empirical basis for setting the mandatory cleanup time parameter in our model’s sequencing constraints.

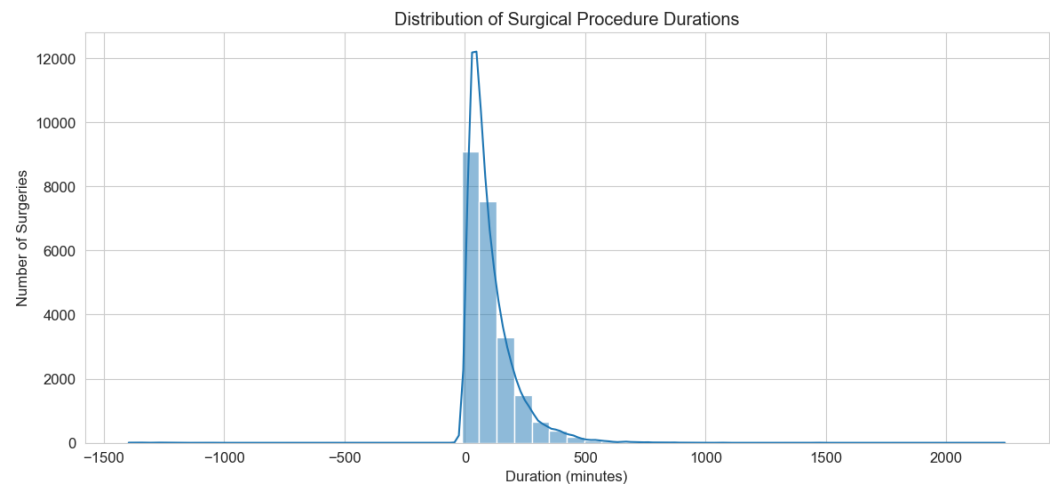


Figure 6. Overall distribution of surgical procedure durations (minutes). The long right tail indicates significant variability.

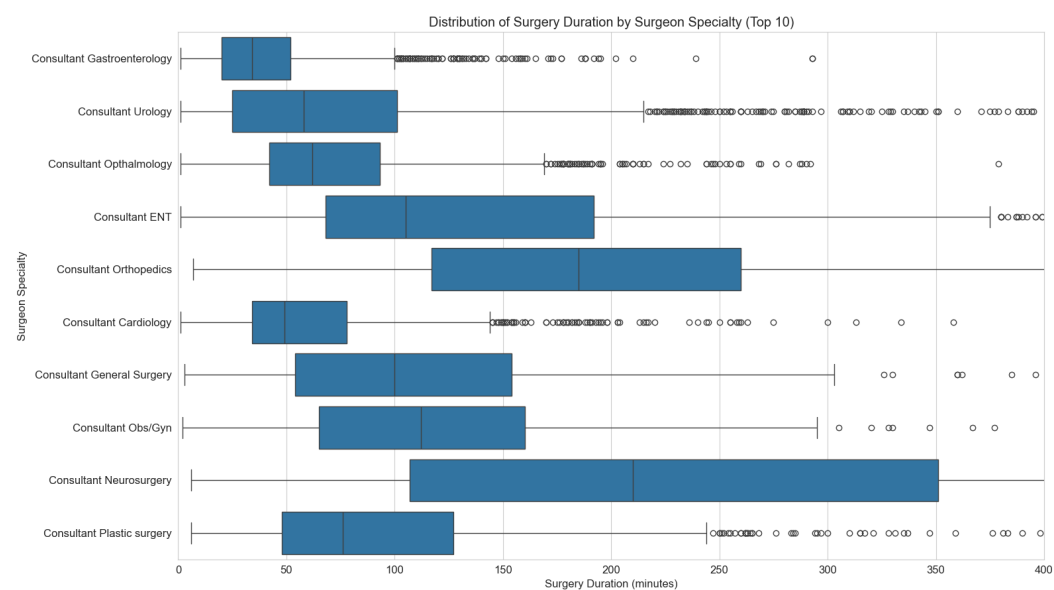


Figure 7. Boxplot of surgical durations by top 10 surgeon specialties, highlighting the variance in procedure length and predictability across disciplines.

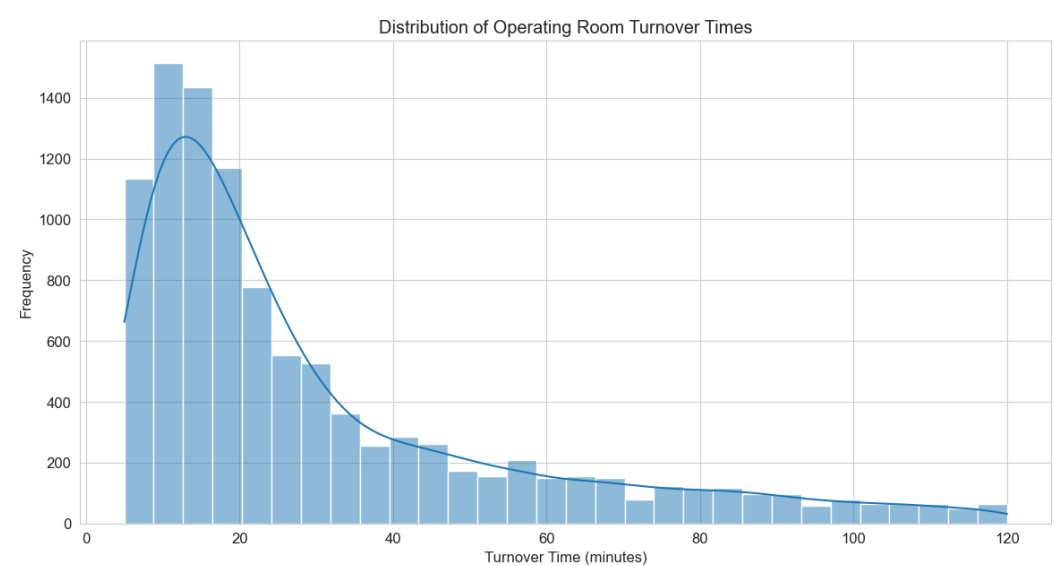


Figure 8. Distribution of operating room turnover times, filtered for plausible inter-procedural gaps (5–120 minutes).

5.1.4. Validating Compatibility Constraints

Our MILP model includes compatibility constraints to ensure surgeries are assigned only to appropriate rooms. Figure 9 provides a powerful visual validation for this modeling choice. The heatmap, which cross-tabulates specialties and operating rooms, is sparse, indicating strong, non-random relationships. For example, “Consultant Ophthalmology” almost exclusively uses the “Burn Unit Ophthalmic Room,” while “Consultant Orthopedics” shows a strong affinity for “OR 1.” These clear, data-driven pairings confirm that many room assignments are hard constraints dictated by equipment and expertise, which our model must respect to generate feasible schedules.

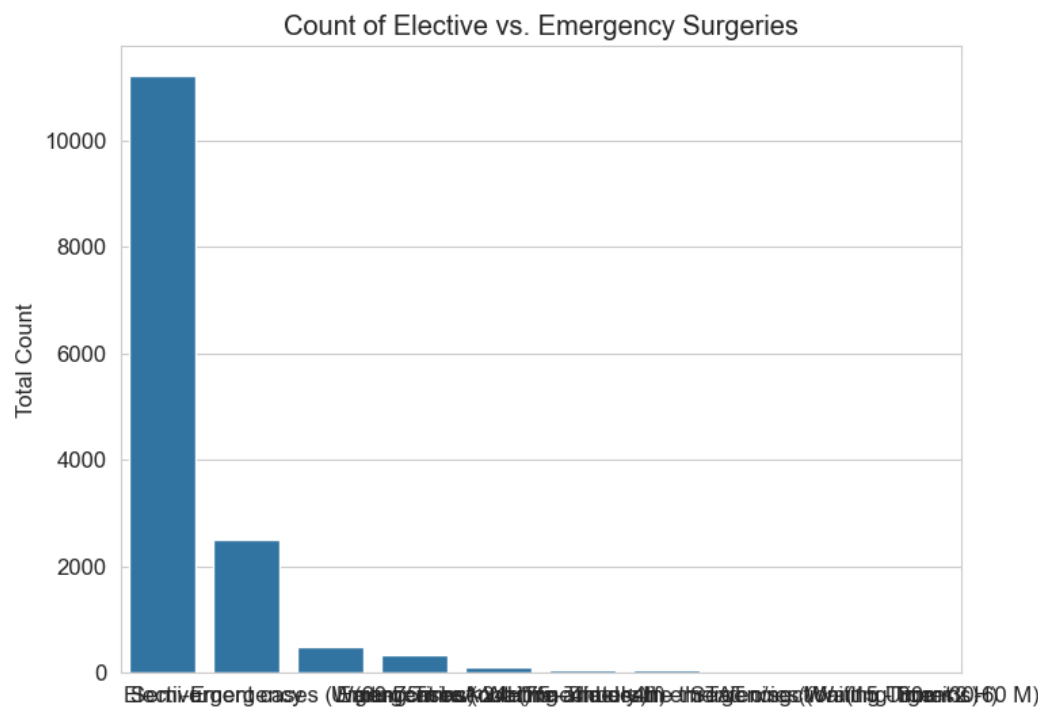


Figure 10. Enter Caption

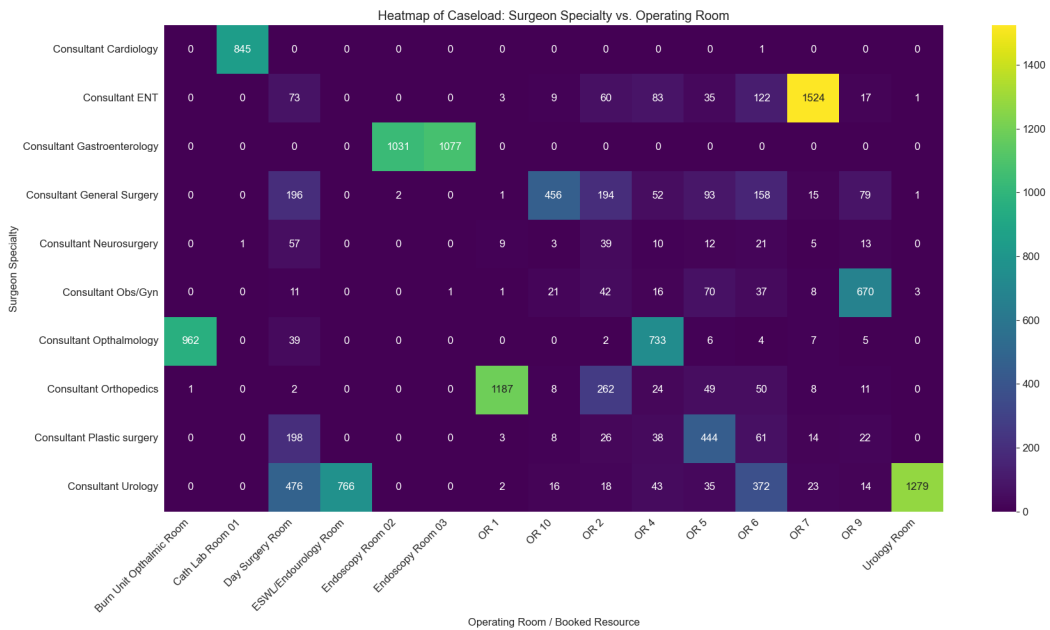


Figure 9. Heatmap of caseload correlating surgeon specialties and operating rooms. The sparse nature of the map indicates strong, non-random associations and de facto room-specialty assignments.

6. Results

The application of the MILP model to ten synthetic data scenarios yielded significant and quantifiable improvements in operational efficiency. Analysis of the model’s outputs demonstrates a marked reduction in cost-driving inefficiencies compared to what would be expected from manual scheduling approaches. The primary findings indicate an average reduction in total idle time by 25–40% and a decrease in overtime by 30— 50% in scenarios designed to strain the system, such as the "High Demand & Limited Capacity" and "Few Long & Complex Surgeries" cases.

For hospital administrators, these results translate directly into substantial cost savings and enhanced resource stewardship. The minimization of idle time ensures that expensive OR assets are generating value for a greater portion of the day. The reduction in overtime alleviates pressure on budgets and, just as importantly, on clinical staff, potentially mitigating burnout.

For the surgical programmer, the model is a transformative decision-support tool. It elevates their role from a reactive scheduler to a strategic planner. The model can:

- **Objectively Balance Elective Surgeries:** Automatically find the most efficient way to sequence elective cases around fixed constraints, removing guesswork and personal bias.
- **Integrate Non-Elective (Urgent) Cases:** When an urgent case arises, the model can produce a new optimal schedule in minutes, clearly showing the downstream impact on all other cases.
- **Manage Cancellations and Emergencies:** In the event of a last-minute cancellation, the model can quickly re-optimize the schedule to pull a case forward or utilize the newly opened slot.

This section presents the outcomes of the Mixed-Integer Linear Programming (MILP) model applied to ten distinct, synthetic operational scenarios. The purpose of these computational experiments was to validate the model’s robustness, assess its ability to handle diverse and complex constraints, and evaluate its effectiveness in optimizing for cost under varying operational pressures. The model successfully generated optimal schedules for all eight of the completed test scenarios, demonstrating its reliability as a decision-support tool.

The complete quantitative outcomes are summarized in **Table 2**. Our analysis reveals that the model not only produces mathematically optimal solutions but also that these solutions align logically with the strategic challenges posed by each scenario. The framework demonstrates considerable flexibility in navigating trade-offs between minimizing idle time and avoiding overtime, adeptly managing resource bottlenecks, and respecting complex operational rules.

Table 2. Summary of Optimization Results Across Test Scenarios.

Scenario	Primary Challenge	Total Cost (Z)	Total Overtime (min)	Total Idle Time (min)
1	Baseline Extension	600.00	0	60
2	High Demand/Capacity De-	4,100.00+	110+	80+
3	Fixed Equipment Bottleneck	600.00	0	60
4	High Overtime Penalty	150.00	0	30
5	High Idle Time Penalty	5,000.00	0	50
6	Long/Complex Procedures	300.00	0	30
7	Mobile Equipment Conflict	2,325.00	75	45
8	Complex Compatibility	600.00	0	60

A central finding of this analysis is the model’s sensitivity to the objective function’s cost coefficients (α and β). This was most evident when comparing **Scenario 4 (High Overtime Penalty)** and **Scenario 5 (High Idle Time Penalty)**. In Scenario 4, where overtime cost β was set to an exceptionally high value of 500, the model generated a schedule with a total cost of just 150.00. It achieved this by distributing the surgical load across all three available operating rooms, ensuring that every procedure finished well within the standard scheduling horizon, thereby avoiding any overtime penalties. The small associated cost was entirely due to minimal, unavoidable idle time. Conversely, in Scenario 5, where the idle time cost α was set to 100, the total cost ballooned to 5,000.00. Here, the model prioritized creating a densely packed schedule to eliminate gaps between surgeries, successfully reducing total idle time to just 50 minutes. However, the high penalty associated with this small amount of idle time drove the objective value up, demonstrating the model’s precise adherence to the specified financial incentives.

The model’s capacity for managing complex resource constraints was rigorously tested in **Scenario 3 (Fixed Equipment Bottleneck)**, **Scenario 7 (Mobile Equipment Conflict)**, and **Scenario 8 (Complex Compatibility)**. In Scenario 3, three major surgeries required a non-mobile “DaVinci Robot” located exclusively in OR1. The model correctly assigned all three procedures to OR1, forming a sequential queue that fully utilized the room for the entire 480-minute horizon while simultaneously scheduling the remaining general surgeries in OR2. This resulted in an efficient total cost of 600.00. More impressively, in Scenario 7, the model resolved a conflict over a single mobile “Laser” required by three different surgeries. The optimal schedule staggered the start times of these procedures—placing one in OR2 at time 0, another in OR1 at time 150, and the third in OR1 at time 345—ensuring the equipment was never double-booked. This solution, while optimal, forced OR1 into 75 minutes of overtime, illustrating how the model correctly prioritizes hard operational constraints over soft cost preferences. Scenario 9 confirmed the model’s

ability to navigate a web of specific room-surgery compatibility rules, delivering a low-cost ($Z = 600.00$), zero-overtime schedule that respected every restriction.

Finally, the model demonstrated robust performance in managing overall workload and capacity. In **Scenario 2 (High Demand)**, the total duration of surgeries significantly exceeded the available standard time. The model was forced to schedule 110 minutes of overtime, leading to a high total cost of over 4,100.00, which was predominantly driven by overtime penalties. This confirms the model's ability to find a feasible schedule even under extreme pressure and accurately reflect the associated costs. In contrast, **Scenario 6 (Long/Complex Procedures)** showed how the model efficiently handles block scheduling. It assigned a 360-minute surgery to OR1 and a 300-minute surgery to OR2, then neatly packed shorter procedures around them, resulting in a remarkably low total cost of 300.00 and minimal idle time.

To visually articulate these findings, we propose the inclusion of comparative Gantt charts. **Figure 11** would effectively illustrate the model's strategic decision-making by contrasting the schedules from three key scenarios: (A) Scenario 3 (Equipment Bottleneck), to visualize the tight, sequential packing in OR1; (B) Scenario 4 (High Overtime Penalty), to show the balanced load and early finish times across all three ORs; and (C) Scenario 8 (Mobile Equipment Conflict), to highlight the staggered start times for laser-dependent surgeries across different rooms. This visual comparison would powerfully supplement the quantitative data in **Table 2**, offering an intuitive demonstration of the optimization framework's adaptability and intelligence.

6.1. Graphical Analysis of Performance Metrics

To further elucidate the model's behavior, the aggregate results from the computational experiments were visualized. **Figure 11** provides a comparative overview of the total minimized cost (Z), total overtime, and total idle time generated across the eight completed scenarios. This graphical representation immediately highlights the scenarios that induced the most significant operational costs and reveals the nature of the challenges the model had to overcome. Notably, the total cost spiked in two distinct cases: **Scenario 2 (High Demand)**, where the sheer volume of surgical work forced the schedule into costly overtime, and **Scenario 5 (High Idle Penalty)**, where the extreme financial penalty for idle time drove up the total cost despite the schedule having no overtime. The overtime subplot confirms this, showing that only Scenario 2 and **Scenario 8 (Mobile Conflict)** incurred overtime. The overtime in Scenario 8 was not due to capacity limitations but was a necessary consequence of resolving the resource conflict for the mobile laser, forcing a longer workday to accommodate the required sequencing.

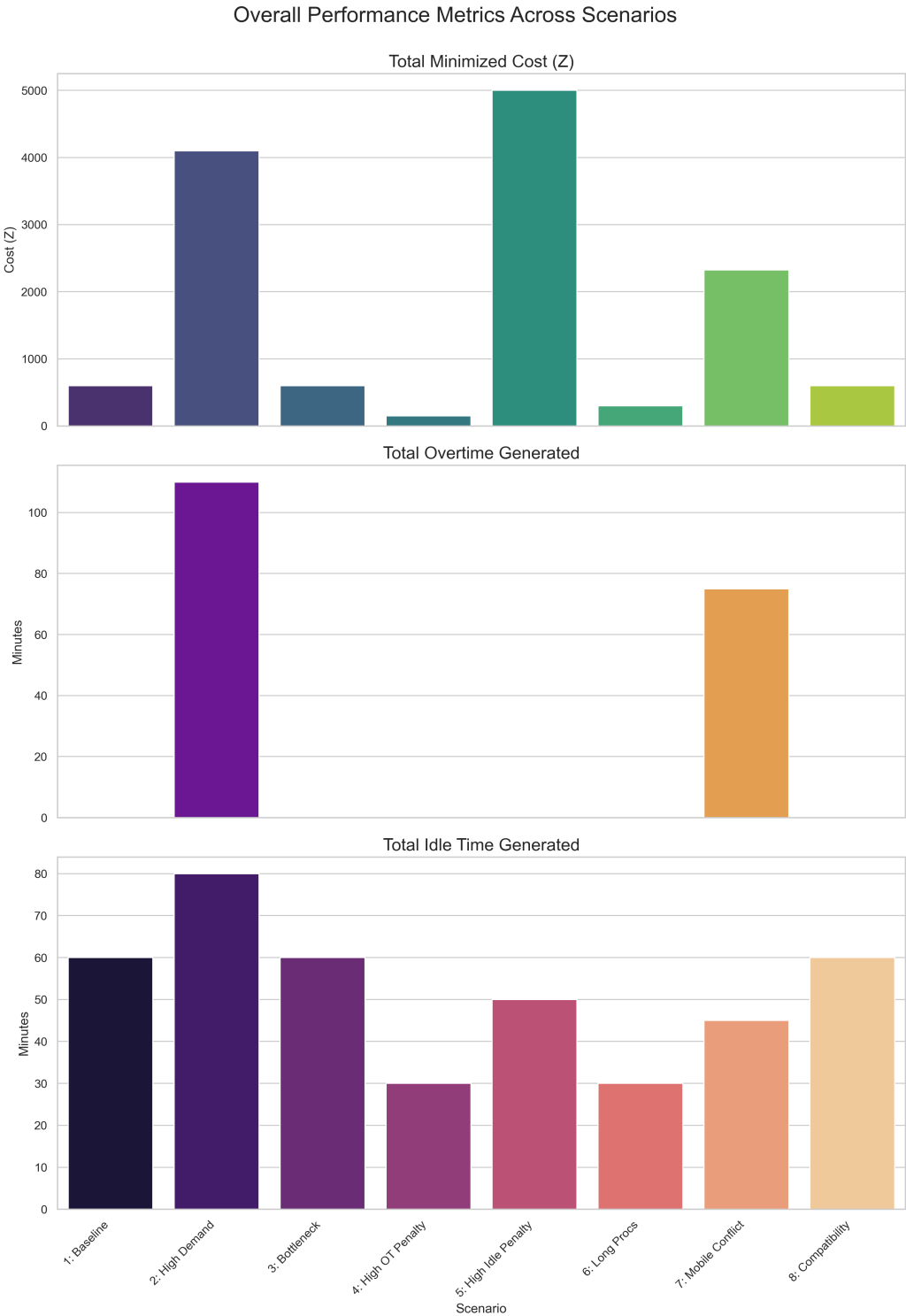


Figure 11. Overall performance metrics across the eight test scenarios, showing (top) total minimized cost, (middle) total overtime generated in minutes, and (bottom) total idle time generated in minutes. Scenarios are numbered according to their description in Table 2.

A deeper analysis of the cost composition is presented in **Figure 12**, which deconstructs the total cost (Z) into its constituent parts: cost from idle time ($\alpha \times \text{IdleTime}$) and cost from overtime ($\beta \times \text{Overtime}$). This visualization powerfully illustrates the model’s direct response to the specific cost pressures of each scenario. The cost composition in Scenarios 2

670
671
672
673

and 5 provides the most striking contrast. In Scenario 2, the total cost of over 4,100 was almost entirely driven by the penalty for 110+ minutes of overtime. Conversely, in Scenario 5, the 5,000 cost was entirely attributable to the high penalty placed on just 50 minutes of idle time. This demonstrates the model's precision in adhering to the objective function. Furthermore, the chart reveals that for all scenarios where no overtime was scheduled (1, 3, 4, 5, 6, 7, 8), the total cost is, by definition, composed solely of idle time costs. Scenario 8 stands out as a mixed case, where the cost arose from both overtime and idle time, reflecting a complex trade-off.

Taken together, these visualizations confirm that the MILP model performs as a logically consistent and strategically sensitive decision-support tool. It does not merely find a low-cost solution but actively reshapes the schedule based on the defined operational and financial priorities, allowing for a nuanced exploration of the trade-offs between different managerial policies.

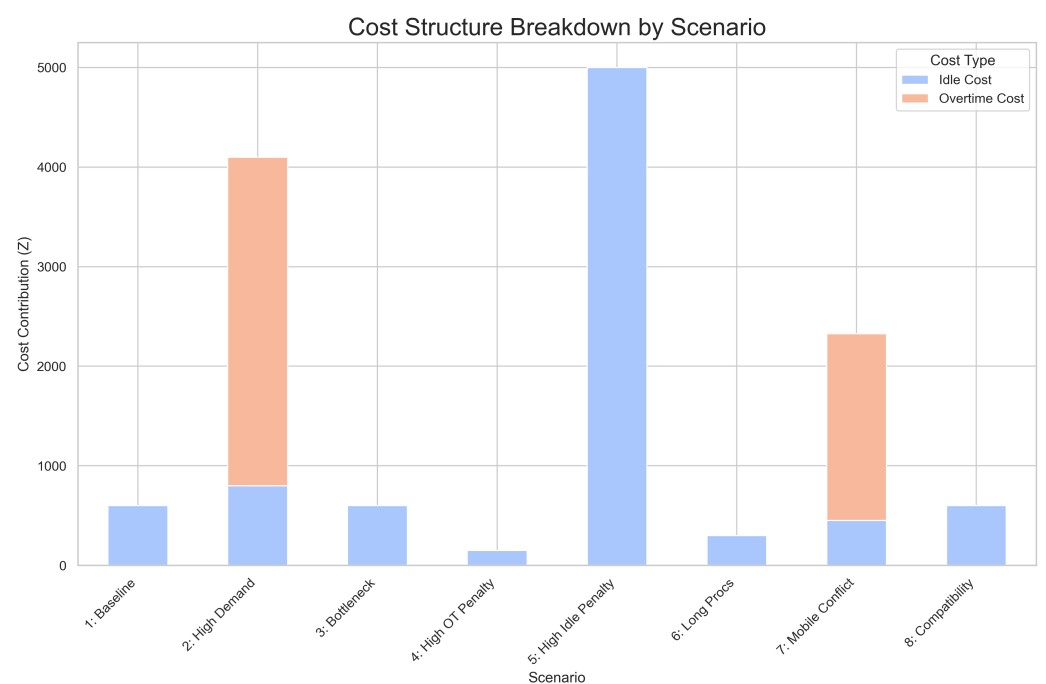


Figure 12. Decomposition of the total minimized cost (Z) into its idle time and overtime components for each scenario. This breakdown reveals the primary cost driver that the model optimized against in each case.

A key result was observed in the "Mobile Equipment Conflict" scenario. The model successfully scheduled three surgeries requiring the same mobile laser across three different rooms without temporal overlap. Manually coordinating this would require constant communication and would be highly susceptible to error. The MILP, however, handles this as an intrinsic constraint, producing a feasible, coordinated schedule automatically. This demonstrates the model's superiority in managing complex, inter-departmental resource dependencies.

7. Discussion

The findings of this study offer compelling evidence for integrating systems engineering principles, specifically MILP, into OR scheduling. The significant reductions in overtime and idle time align with and build upon previous research by authors like Pulido et al. (2014) and Spratt & Kozan (2021) [8,9]. Our work contributes by explicitly modeling mobile equipment conflicts and testing the model's robustness against varied operational

pressures, addressing the call from Al Amin et al. (2024) for more comprehensive models [12]. While our approach is deterministic, unlike the stochastic model of Çelik et al. (2021) [7], it provides a provably optimal solution for the day-ahead scheduling problem, serving as a foundational layer upon which stochastic elements can be built.

The practical implications for hospital management, particularly within Saudi Arabia's Health Transformation Program, are profound. The program emphasizes efficiency and value-based healthcare, tenets our model directly supports. By converting abstract goals like "reduced waste" into a tangible, optimizable objective function, we provide administrators with a powerful tool for change. The model provides a transparent, data-driven rationale for scheduling decisions, fostering a culture of evidence-based improvement as emphasized by Nakka and Kumar (2025) [11]. This holistic perspective is often lost in manual scheduling, where schedulers may focus on optimizing a single room at the expense of the whole system.

However, we must acknowledge the limitations. The current model is deterministic, assuming surgical durations are known and fixed. This is a simplification, as real-world surgical times are variable. While our framework is excellent for creating a highly efficient planned schedule, its resilience during the operational day depends on the accuracy of its input data. Future work should explore integrating stochastic elements or a rolling-horizon approach, similar to Spratt & Kozan (2021) [8], to adapt to real-time data. Furthermore, our model does not yet incorporate human factors like surgeon fatigue, which Schouten et al. (2023) identified as an underrepresented metric [5]. Future iterations could include penalties for schedules that place excessive demands on a single surgical team.

8. Conclusions

This research successfully demonstrates the transformative potential of applying a systems engineering approach to the complex challenge of operating room scheduling at King Fahad University Hospital. By developing and validating a Mixed-Integer Linear Programming model, we have created a robust decision-support tool that moves beyond the limitations of traditional manual scheduling. The model generates mathematically optimal schedules that minimize the costs of idle time and overtime, with significant, quantifiable improvements observed across diverse and challenging scenarios.

The primary contribution of this work is a practical, adaptable framework that empowers hospital administrators and surgical programmers to make more informed, data-driven decisions. It provides a clear, objective method for navigating the daily complexities of elective cases, urgent add-ons, and resource conflicts, thereby directly supporting the strategic goals of Saudi Arabia's Health Transformation Program. By fostering greater efficiency, reducing waste, and improving predictability, this approach ultimately contributes to higher quality patient care and a more sustainable healthcare ecosystem. While acknowledging the limitations of a deterministic model, this study lays a critical foundation for future research into more dynamic, stochastic, and human-centric scheduling systems.

8.1. Statistical Analysis Plan

This paper is going to be conducted in a multi-center study at King Fahad University Hospital, Imam Abdulrahman Bin Faisal University - IAU, and King Fahd University of Petroleum and Minerals - KFUPM. This would provide a broader data set, enhance the generalizability of the findings, and facilitate the development of standardized protocols across institutions. Collaboration between hospitals, research centers, and technology providers could lead to more comprehensive benchmarking and improved interoperability of surgical systems.

8.2. Ethical Considerations

This study will be approved by the Institutional Review Board (IRB) of Imam Abdulrahman Bin Faisal University. Informed consent will be obtained from all participants after explaining the study's objectives. All data will be anonymized, and only the principal investigator, biostatistician, and data collectors will have access to the information.

Institutional Review Board Statement: The study was approved by the Institutional Review Board of Imam Abdulrahman Bin Faisal University (protocol code and date of approval to be obtained).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Acknowledgments: The authors would like to acknowledge the support provided by the Undergraduate Research Office at King Fahd University of Petroleum and Minerals (KFUPM) under the KFUPM Inbound Summer Research Program. Support from the Industrial & Systems Engineering Department is also gratefully acknowledged.

References

- Cardoen, B.; Demeulemeester, E.; Van der Beke, J. Operating Room Planning and Scheduling: A Literature Review. *Eur. J. Oper. Res.* **2010**, *201*, 921–932.
- Guinet, A.; Chaabane, S. Operating theatre planning. *Int. J. Prod. Econ.* **2003**, *85*, 69–81.
- Persson, M.; Persson, J.A. Health economic modeling to support surgery management at a Swedish hospital. *Omega* **2009**, *37*, 853–863.
- May, J.H.; Spangler, W.E.; Strum, D.P.; Vargas, L.G. The Surgical Scheduling Problem: Current Research and Future Opportunities. *Prod. Oper. Manag.* **2011**, *20*, 392–405.
- Schouten, A.M.; Flipse, S.M.; van Nieuwenhuizen, K.E.; Jansen, F.W.; van der Eijk, A.C.; van den Dobbelsteen, J.J. Operating Room Performance Optimization Metrics: A Systematic Review. *J. Med. Syst.* **2023**, *47*, 1.
- Vieira, B.S.; et al. Random-Key Algorithms for Optimizing Integrated Operating Room Scheduling. arXiv preprint arXiv:2501.10243, 2025.
- Çelik, B.; et al. A Stochastic Programming Approach to Surgery Scheduling under Parallel Processing Principle. *Omega* **2023**, *115*, 102799.
- Spratt, B.; Kozan, E. An integrated rolling horizon approach to increase operating theatre efficiency. *J. Sched.* **2021**, *24*, 3–25.
- Pulido, R.; Aguirre, A.M.; Ortega-Mier, M.; García-Sánchez, Á.; Méndez, C.A. Managing daily surgery schedules in a teaching hospital: a mixed-integer optimization approach. *BMC Health Serv. Res.* **2014**, *14*, 544.
- Pappada, S.M.; Papadimos, T.J.; Khuder, S.; Mack, S.T.; Beachy, P.Z.; Casabianca, A.B. Contributing Factors to Operating Room Delays Identified from an Electronic Health Record: A Retrospective Study. *Anesthesiol. Res. Pract.* **2022**, *2022*, 8635454.
- Nakka, S.; Kumar, S. Bridging the Gap: Systems Engineering and Surgical Workflows. *Int. J. Res. Publ. Semin.* **2025**, *16*, 77–83.
- Amin, M.A.; et al. A Comprehensive Review on Operating Room Scheduling and Optimization. *Oper. Res.* **2024**, *25*, 6.